

Database

↳ Structured and organized set of data

Relational (RDBMS)

↳ MySQL, PostgreSQL,
Oracle

NoSQL //

⇒ DBMS ⇒ Software app to manage and manipulate data.

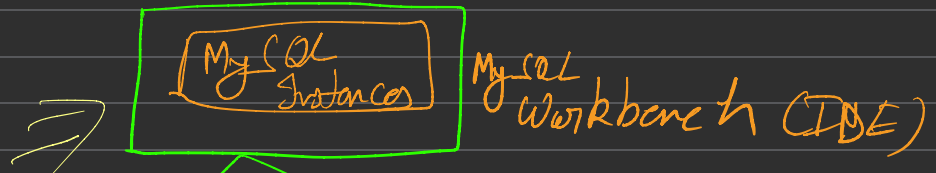
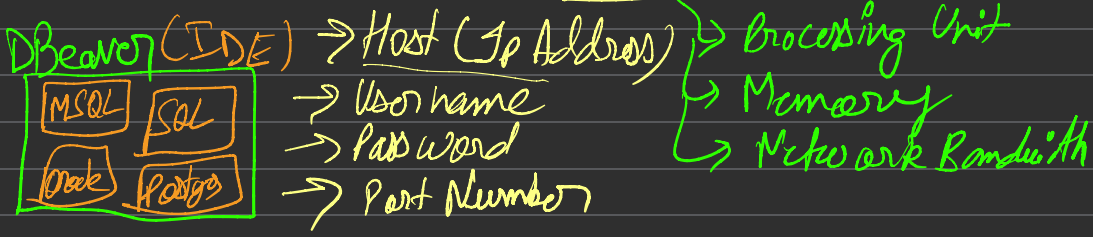
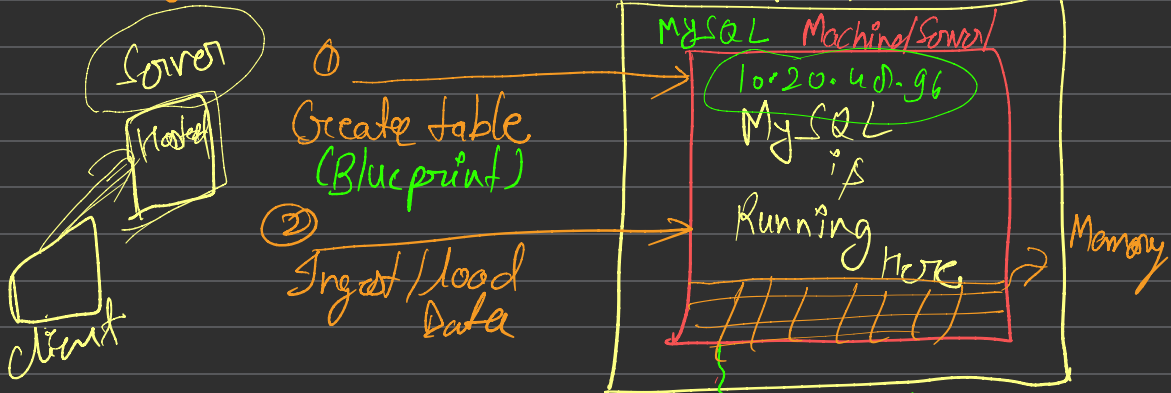
⇒ RDBMS ⇒ Software app to manage data, structured organized data in tables form which may link with each other based on keys.

MySQL

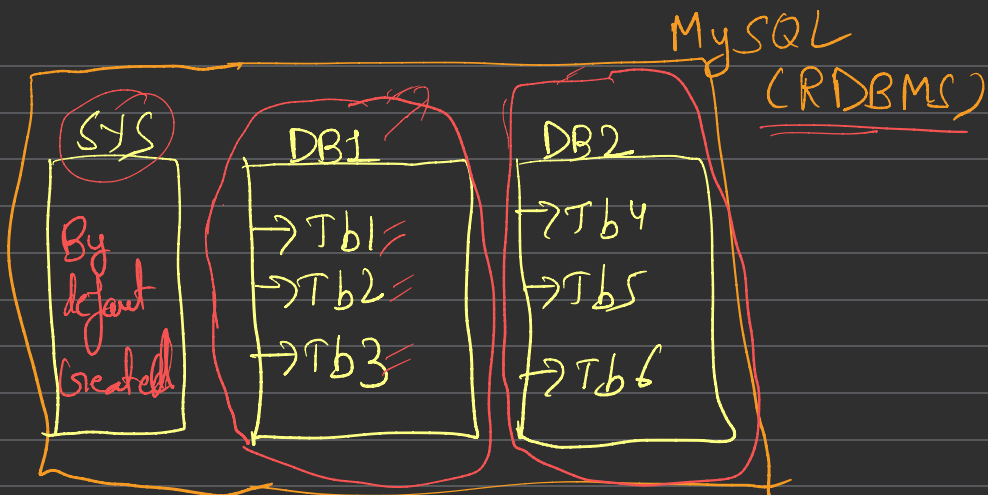
employee			department	
empid	name	dept_id	deptid	dept name
1	A	100	100	IT
2	B	100	200	HR
3	C	200	300	Finance

Setup

MySQL $\Rightarrow$ RDBMS

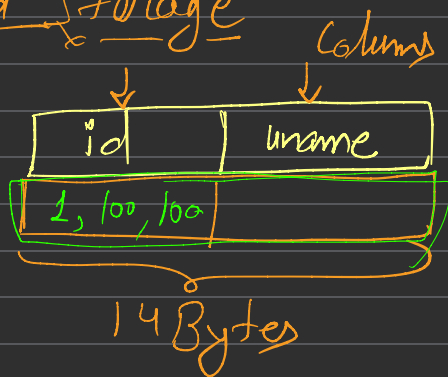


$\rightarrow$ Host (localhost) $\Rightarrow$ 127.0.0.1, 0.0.0.0



Row based storage

Create table words
 {
 id int,
 unname varchar(10)
 };



int $\Rightarrow$ 4 Bytes

char $\Rightarrow$ 1 Byte

varchar(10) = 10 Bytes

id $\Rightarrow$ 100
 unname $\Rightarrow$ 'Amit' } Row

100 | Amit

Row 1 How much memory? 14 Bytes

Row1

100 101 102 103 104 105 106 107 108 109 110 111 112 113

Integer 10 Bytes for Username Varchar

Row2

Integer

10 Bytes for Username

114 115 116 117 118 119 120 121 122 123 124 125 126 127

Memory Area